

Beyond Subjects: Topic Situations, Multiple Agree, and the Diachrony of Switch-Reference
Morphology

Enrico Higginbotham
Advisor: Karlos Arregi
Preceptor: Alexandra Fraser

THESIS SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR
THE DEGREE OF MASTER OF ARTS

MASTER OF ARTS PROGRAM IN THE HUMANITIES
UNIVERSITY OF CHICAGO

19 MAY 2025

1. Introduction

In this work, I propose an account of switch reference. This analysis relies on a parametrized Multiple-Agree probe in a specialized functional projection. Depending on the parameter setting, the probe copies either the referential index values of subjects or topic situation pronouns. I also demonstrate how this mechanism contributes to accounting for the diachronic evolution of switch-reference morphology. Switch reference is a morphosyntactic mechanism that marks whether the subjects of linked clauses are identical or distinct (Baker & Camargo Souza 2020:3). Canonically, switch reference surfaces as a verbal suffix with specialized same-subject and different-subject morphemes, tracking subject identity across clauses. As an illustrative example, consider the following minimal pair in Yavapai (Upland Yuman, Arizona), where the morpheme *-k* marks coreference between the subject of *u* ‘see’ and *čikwar* ‘laugh’ (both being *Tokatoka*) and the morpheme *-m* marks disjoint reference between the subjects of the same two verbs (one being *Tokatoka* and the other being *Savakyuva*):

Yavapai

(1) a. *tokatoka-č savakyuva u-t-k čikwar-kiñ.*

Tokatoka-SU Savakyuva_k see-TEMP-SS laugh-COMP

‘When Tokatoka_i looked at Savakyuvak, he_i laughed.’

b. *tokatoka-č savakyuva u-t-m čikwar-kiñ.*

Tokatoka-SU Savakyuva see-TEMP-DS laugh-COMP

‘When Tokatoka_i looked at Savakyuvak, he_k laughed.’ (Roberts 2017:1)¹

¹ The abbreviations used in this work are: 1 = first person, 2 = second person, 3 = third person, ABS = absolutive case, ABS.FORM = absolute form, ACC = accusative case, ALLAT = allative, AUX = auxiliary, CAUS = causative, CMP = completive, c(OMP) = complementizer, DECL = declarative, DET = determiner, DIST = distal, DS = different subject/situation, DUB = dubitative, DUR = durative, ERG = ergative case, FUT = future tense, IMP = imperative, IMPF = imperfect tense, INC = inceptive, IND = indicative, INDF = indefinite, INS = instrumental, INV = inverse number, IPFV = imperfective aspect, i = inclusive person, IRR = irrealis, LOC = locative, MED = medial, NEG = negative, NOM = nominative, OBJ = object, OBL = oblique, PERF = perfect tense, PF/PFV = perfective aspect, PL = plural, PRED = predicate marker, PRO = pronoun, PROSP = prospective aspect, PRS = present tense, PRT = particle, PST = past tense, px = exclusive person, Q = question marker, QUOT = quotative, REFL = reflexive, REM = remote past tense, SBJ/SU = subject, SEQ = sequential, SG = singular, SIMUL = simultaneous, SS = same subject/situation, S = sentence, ST = stative, TEMP = temporal, TOP = topic, TR = transitive

In its original identification as a functional category in Washo, switch reference strictly tracks grammatical subjects and is closely integrated with clausal morphology. Therefore, it offers a fruitful line of inquiry for theories addressing cross-clausal relations and reference tracking. However, in the last 50 years, researchers have documented systems of non-canonical switch reference in which the relevant morphology extends beyond simple subject tracking, tracking instead situational shifts, narrative continuity, and discourse conditions such as topic continuity or breaks in narrative coherence (Haiman & Munro 1983; Stirling 1993; de Sousa 2016), as seen in (2) in Amele (Papuan, Papua New Guinea). In this example, the different-subject morpheme surfaces to mark a change of place, despite the same subjects between the two clauses (Stirling 1993:216):

Amele

(2) *Age ceta gul-do-co-bil ʔ-i bahim na tac-ein*

3PL yam carry-3SG-DS-3PL go-PRED floor on fill-3PL:REMP

‘They carried the yams on their shoulders and went and filled up the yam store.’

(Stirling 1993:216)

These patterns of non-canonical switch-reference are challenging for current theoretical accounts of switch reference and raise the fundamental issue of whether they constitute extensions of canonical switch reference or represent a separate phenomenon altogether.

The Uto-Aztecan language family offers particularly insightful and varied empirical evidence for this issue. Uto-Aztecan languages encompass both robust canonical switch-reference systems, as found in Hopi, and non-canonical switch-reference systems, such as that seen in Serrano. Hopi exemplifies canonical switch-reference, strictly tracking grammatical subjects. Serrano, on the other hand, employs non-canonical switch-reference markers to indicate the relation between situations described by different clauses, rather than strictly tracking subject identity. Tohono O’odham represents another interesting variant, having historically inherited

switch-reference morphology that evolved into more general coordinating conjunctions between specific syntactic categories.

Drawing from Arregi & Hanink (2022) and McKenzie (2012), I argue for a unified, parametric analysis of switch-reference morphology through the syntactic operation of Multiple Agree, where a probe bearing an [ID] feature in a specialized switch-reference head copies the referential index value of either subjects or topic situation pronouns. I propose that the distinction between canonical and non-canonical switch reference in non-intensional contexts (i.e., coordination and clause chaining) results precisely from this parametric option. By applying this parametric Multiple Agree framework to Hopi, Serrano, and Tohono O’odham, this analysis demonstrates how minimal parametric variations can systematically derive canonical and non-canonical switch reference, and account for historical developments leading from morphological switch-reference markers to more generalized coordinating conjunctions. This diachronic perspective further underscores the explanatory power of the parametric model, illustrating how the interaction between different syntactic configurations and parameter settings over time can yield significant morphosyntactic and functional changes within closely related languages.

In the following sections, I discuss the relevant typological variation in switch-reference marking, my theoretical contribution to its analysis, and the major accounts that have informed this work. I then proceed to apply the proposed framework to Hopi and Serrano, two Uto-Aztecan languages exhibiting canonical and non-canonical switch reference, respectively. Finally, I analyze the diachrony of Uto-Aztecan switch-reference morphology in Tohono O’odham and offer an account of its historical trajectory through the principle of Feature Economy.

2. Sources

All linguistic examples cited in this work are drawn exclusively from published or previously archived sources; no new fieldwork was undertaken. Hopi examples are drawn from Jeanne's (1978) dissertation text, Hale (1992), and Stahlschmidt (1983). Malotki (1983) was consulted specifically to address whether Hopi exhibits non-canonical switch-reference patterns. Serrano examples primarily derive, through Hill & Hill (2019) and Hill (2016), from *Wayta' Yawa'* 'Always Believe,' a bilingual text born from the collaboration of linguist Dr. Eric Elliott with Mrs. Dorothy Ramón, recounting her life on the Morongo Indian Reservation. Additional Serrano examples not originally recorded in *Wayta' Yawa'* were drawn from Hill (2016) and Hill & Hill (2019). The Tohono O'odham data are primarily excerpted from Zepeda's (1983) reference grammar and Mathiot's (1974) dictionary, with additional examples from Hale (1983). Further insight into Tohono O'odham derives from my experience as a Tohono O'odham language learner at the University of Arizona and personal communication with Dr. Jeremy Johns. Illustrative examples from Yavapai, Cupeño, Kiowa, and other languages are cited verbatim from the works listed in the References. All remaining errors are my own. The resources on Uto-Aztecan languages consulted in this work are available through the Uto-Aztecan Linguistics Archive, hosted on Zotero (https://www.zotero.org/groups/5669330/uto-aztecan_linguistics_archive/library).

3. Canonical and non-canonical switch reference

Switch reference is traditionally understood as a morphological mechanism that marks whether the subjects of linked clauses are the same or different (Baker & Camargo Souza 2020:3). The clause marked with switch-reference morphology is termed the *pivot clause*, and the one without switch-reference morphology is termed the *anti-pivot clause*. Under a *canonical* switch reference system, the morphology straightforwardly and uniquely tracks subject co-reference: if the embedded clause and matrix clause share a subject, one morphological exponent (i.e., *same-subject*) is realized; if they have different subjects, another exponent (i.e., *different-*

subject) surfaces. As seen in (1) in Yavapai, this mechanism is also frequent in Uto-Aztecan languages, both Northern and Southern, as can be seen in (3) in Cupeño (Northern Uto-Aztecan) and in (4) in Huichol (Southern Uto-Aztecan):

Cupeño

- (3) *Mu=ku'ut* [*pi='ami-nuk*], *pi=pe-meq*
 and=QUOT [*him=knock.down-(TR)-SS*] *him=he-kill:SG:PST*
 'And it is said that having knocked him down, he killed him.' (Hill 2016:129)

Huichol

- (4) *ʔuuka nua-ku, nee ne-petia.*
 Girl arrive-DS 1SG 1SG-leave
 'When the girl arrived, I left.' (Comrie 1983:19)

Although this variant of switch reference holds for a diverse set of languages, many others display *non-canonical* switch reference, in which morphological marking does not appear to track nominal reference. Instead, the presence of same-subject or different-subject marking is governed by the perceived natural unity or disunity of the predications in the relevant clauses, whether along temporal, causal, or attitudinal dimensions. In these cases, I call the relevant morphology *same-situation* and *different-situation*. This instantiation of switch-reference phenomena is exemplified in Northern Uto-Aztecan by Serrano and Cahuilla, where non-canonical switch reference is employed to indicate the degree of connection between the situations of the two clauses. In the Cahuilla example in (5), different-situation marking is used in a same-subject context as a source of situation discontinuity (Hill 2016:137). Conversely, in the Serrano example in (6), same-situation marking is employed as a source of continuity between the situations presented in the pivot and anti-pivot clauses:

Cahuilla

- (5) [*Pe'-ish pe' pe'-iy ne' tax-ne-ting'ay-qal-ipa'*
 it-INS TOP DET-ACC I REFL-I-medicate IMPF:SG-DS
samat p-ish] *pepiy ne-hichi-qa pen-'ayi-k.*
 herb it-INS] far I-go-PRS:SG 1SG>3SG-gather-INC
 'That is why when I need to treat myself with herbs I go far out into the wilderness.'

(Hill 2016:137)

Serrano

(6) *Taaqtam herheermim nyiivayu*, ‘ama’ meempt kwenevu’ ‘apyu’ werra’n, chaacu’

(R&E 213)²

[*Taaqt-am ho’hoo’-m-im ñi-ivaju*’], ama’ miimt kwini=vi’ apiu’

[person-PL shamans-PL do-ss] that ocean QUOT=3SG>3SG from.there

vira’n, čaaçu’

speak sing

‘If you are a shaman, the ocean would reportedly speak or sing (to you) from there;
literally: ‘When people are shamans, that ocean can speak to him from there, (or) sing.’

(Hill 2016:120)

Although the labels of “canonical” and “non-canonical” are standardly employed in the literature to distinguish between two different manifestations of switch-reference phenomena, they merely name two variants; “non-canonical” does not mean that the pattern is exceptional or accidental. These labels are the result of historical happenstance. William H. Jacobsen Jr. first coined the term in 1967 to unify a series of observations in Washo, which exclusively exhibits canonical switch reference (Arregi & Hanink 2022). Empirical observations in the 1980s and 1990s on a variety of switch-reference-bearing languages (e.g., Haiman & Munro (1983); Stirling (1993)) demonstrated that switch reference could not be systematically limited to subject co-reference without admitting an unacceptable level of unexplained exceptions that showcased a high degree of systematicity. Therefore, the distinction between canonical and non-canonical switch reference was drawn with respect to Jacobsen’s initial investigation and is adopted to this day to distinguish between the two occurrences of the same underlying mechanism.

4. The parametric proposal

² Serrano examples drawn from *Wayta’ Yawa’* ‘Always Believe’ are referenced by the first initial of the authors (Ramón and Elliott) and the page number of the 1st edition published by Malki-Ballena Press.

I propose that the morphological exponent in switch reference arises through a Multiple Agree mechanism hosted by a dedicated switch-reference projection (i.e., SRP) headed by SR^0 . More specifically, SR^0 bears an [ID] feature and probes both downward and upward (i.e., in the pivot and the anti-pivot) to copy the referential index values of subjects in the case of canonical switch reference or of topic situation pronouns in the case of non-canonical switch reference. If the copied values of the [ID] features are identical, same-subject or same-situation morphology is inserted. If the copied values are distinct, different-subject or different-situation morphology is inserted.

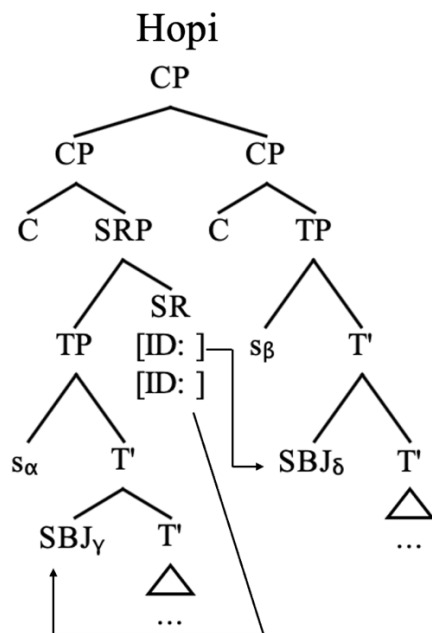
My analysis unifies insights from McKenzie (2012) and Arregi & Hanink (2022) to explain a wider range of cross-linguistic data on non-canonical switch-reference and to demonstrate how a feature-driven, Agree-based analysis of switch reference can account for both synchronic cross-linguistic variation and the diachronic evolution of switch-reference morphology. From McKenzie (2012), I adopt the crucial assumption that non-intensional clauses contain a null topic situation pronoun *s* in the derivation. At the same time, I concur with Arregi & Hanink (2022) by rejecting the assumption that SR^0 intrinsically bears a [DS] or [SS] feature. I adopt their conclusion that the head bearing switch-reference morphology probes both downward and upward for the referential index value of nominative subjects. I argue, however, that cross-linguistic variation between canonical and non-canonical switch reference is parametrized based on which element the probe in SR^0 targets in non-intensional contexts (i.e., subjects or topic situation pronouns). Thus, although non-canonical switch reference is necessarily dependent on the presence of a topic situation pronoun in both clauses, it is not necessarily the case that every language will license non-canonical switch reference in non-intensional contexts (*contra* McKenzie (2012)).

Specifically, in languages that license non-canonical switch reference in non-intensional contexts, SR^0 ignores nominal arguments (by Minimal Search) and exclusively targets topic

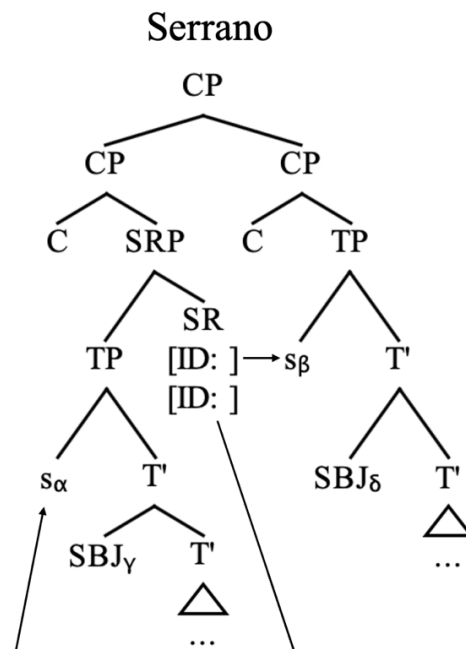
situation pronouns. First, it copies the referential index value of the pivot's topic situation pronoun. Then, it copies the index of the anti-pivot's topic situation pronoun. If both indices match, same-situation morphology appears. If they differ, different-situation morphology appears. By contrast, the probe in SR^0 exclusively targets nominal arguments (i.e., subjects) instead of topic-situation pronouns in languages that only have canonical switch reference. My proposal relies on two Uto-Aztecan languages, Serrano and Hopi, to argue for this Multiple-Agree-based parametric view of switch reference. Hopi exclusively exhibits canonical switch reference, while Serrano also features non-canonical switch reference. Based on the present discussion of parametric variation in non-intensional contexts, I propose the structure and Agree mechanism in (7a) for Hopi and in (7b) for Serrano:

(7)

a.



b.



I also argue that this framework illuminates the attested diachronic paths of switch-reference morphology, where different-subject morphology becomes increasingly independent of the verbal complex of the pivot clause, developing into coordinating morphology lacking an [ID] feature, and same-subject morphology remains associated with the verbal complex and

subclausal coordination. Tohono O’odham exemplifies this process: Proto-Uto-Aztecan different-subject morphemes have undergone phonological reduction and now connect distinct clauses as conjunctions, while same-subject morphemes connect sub-clausal constituents and are markers of simultaneity and sequentiality.

Finally, my proposal carries practical implications for language documentation and revitalization. Because non-canonical switch reference frequently appears in Uto-Aztecan narratives yet does not exist in languages like English, adult learners can mistakenly treat it as an anomaly. By recognizing non-canonical switch reference as a regular and predictable component of a language’s syntax and semantics, both linguists and educators can streamline instructional strategies and highlight the systematicity of a phenomenon that might otherwise seem opaque.

In the next section, I present the major claims of McKenzie (2012) and Arregi & Hanink (2022) that inform the present proposal.

5. Theoretical background

In this section, I outline the theoretical accounts of switch reference in McKenzie (2012) and Arregi & Hanink (2022) that underpin the present analysis, highlighting their key contributions and my points of departure in accounting for switch reference in Hopi and Serrano. Specifically, I address how the situation-semantics approach to non-canonical switch reference in Kiowa presented by McKenzie (2012) and the Multiple Agree approach in Arregi & Hanink (2022) to canonical switch reference in Washo jointly provide the foundational insight necessary for a unified parametric account.

5.1 Topic situations and non-canonical switch reference

Theoretical accounts of non-canonical switch reference have relied on scene-tracking, topic continuity and accessibility relations, or agreement between Davidsonian events (Stirling 1993; Ariel 1990). McKenzie (2012), however, argues for a novel view of canonical and non-

canonical switch reference, where non-canonical switch-reference tracks Austinian topic situations. Austinian topic situations are based on a view of situation semantics in which utterances are about particular situations, with the actual world being the limiting case. They are Merged in the derivation as null pronouns. Semantically, the role of Austinian situations is to restrict the evaluation of truth-conditional content to the context of discourse. In this framework, topic situations are situation pronouns in the verb's extended projection restricting the entire proposition through function application, as a proposition can be defined as the characteristic function of a set of situations. The interpretation rule for situation pronouns is given below from McKenzie (2012:139):

(8) **Situation interpretation (pronoun rule):** For any index n and any assignment g :

$$\llbracket s_n \rrbracket^g = g(n) : s$$

For example, the truth-conditional evaluation of the utterance in (9) is limited to situations consistent with the context of discourse. If we were discussing the weather in Tucson last Monday, the truth evaluation is restricted to the set of propositions in a situation s_1 corresponding to Tucson last Monday:

(9) *It rained.*

(10) a. $\llbracket \textit{It rained} \rrbracket = \lambda s. \text{ it rained in } s : \langle s, t \rangle$

b. $\llbracket s_1 \textit{ It rained} \rrbracket^g = [\lambda s. \text{ it rained in } s] (g(1)) : t = \text{It rained in } g(1) = \text{It rained in } s_1$

The fact that natural language can be modeled using situations is made clear by contradiction. For example, the sentence in (11) results in a contradiction because it cannot be the case that it both rained and did not rain in Tucson last Monday. We understand the truth of both those propositions to be evaluated in the same situation s_1 of events occurring in Tucson last Monday:

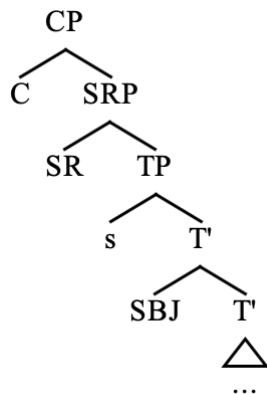
(11) # *It rained in Tucson last Monday and it did not rain in Tucson last Monday.*

Conversely, (12) is fine because it is understood that the first proposition is evaluated in s_1 and the second proposition is evaluated in s_2 (i.e., the situation corresponding to *Tucson last Tuesday*):

(12) *It rained in Tucson last Monday and it did not rain in Tucson last Tuesday.*

The relation between the context of the utterance and its truth-conditional evaluation is relevant to cases of non-canonical switch reference because non-canonical switch reference differs from canonical switch reference in that the relevant morphology tracks the perceived connection between the situations predicated in the two clauses, not merely subject co-reference. In other words, one must look beyond the immediate confines of the clauses' syntactic structure and examine how it is truth-conditionally interpreted in its discourse context. Based on the distribution of complementizers in Kiowa, McKenzie (2012) assumes that the relevant morphology heads its own projection (i.e., SR^0) merged as sister to TP:

(13)



McKenzie (2012) proposes that switch-reference operates differently in intensional and non-intensional subordinate contexts. In his account, a non-intensional subordinate clause (such as a coordinated clause) carries a null topic situation pronoun in Spec, TP, representing the specific situation it describes. The dedicated switch-reference head (SR^0) merged above the

pivot clause's TP uses this situation pronoun to determine whether the two clauses share the same situation.

In a non-intensional context, the switch-reference head first composes with the subordinate topic situation pronoun. Later in the derivation, it composes with the matrix clause's topic situation pronoun. If the two clauses belong to the same situation, the grammar realizes the same-situation form of the switch-reference morpheme. If they belong to different situations, the different-situation form is used. Consider the following context drawn from McKenzie (2012:152):

- (14) *You are talking with your friend about a get-together last weekend. Everyone was supposed to bring something for everyone to eat or drink. You can't remember who it was who brought the coffee and the tea. You know it was two different people, but you can't remember exactly who.*

McKenzie (2012:152) reports that same-situation marking is felicitous in Kiowa when asking who it was that brought the coffee and the tea using a biclausal coordinate structure because the tea- and coffee-bringing both belong to the same situation (i.e., the party):

Kiowa

- (15) *Hâjêl chói bǎu gàu hâjêl chóigùl bǎu?*

hâ:têl tsói Ø- bó: gɔ hâ:têl tsói+gul Ø- bó:
 person\wh coffee [3s:3s] bring:PF **and:SS** person\wh liquid+red [3s:3s] bring:PF

‘Who brought coffee, and who brought tea?’ (McKenzie 2012:153)

Crucially, the speaker commented that different-situation marking would be infelicitous in (15) because “Then it’s like you’re talking about two different parties” (McKenzie 2012:153). Conversely, McKenzie (2012:154) reports that different-situation marking is felicitous when asking what food items have been prepared for two distinct classes on Kiowa culture, each held on a different day, because there is no situational overlap between the two classes, as exemplified by the context in (16) and the sentence in (17):

- (16) *You are teaching a class about Kiowa culture. Each time, you make one type of food. Yesterday it was frybread, today it was boiled meat. I haven't been in your class, so I ask you what you've been making to eat.*

Kiowa

- (17) *Kídèl qáùèjàu ét áúmé nàu éháudèkì císàun é áúmè*
kʰi:dèl kʰʷe:-tɔ ʔéʔ- ʔʷ:m-ě: nɔ é:hɔ:dekʰi
 yesterday fry.bread-INV [1px:3i] eat.PF **and:DS** today
kí+sɔ̃n é- ʔʷ:m-ě:*
 meat+boil [1px:3s] eat:PF
 ‘Yesterday we (excl.) made fry bread and today we made boiled meat.’ (McKenzie 2012:154)

However, in intensional subordinate clauses (e.g., those introduced by an operator such as the conditional *if*), McKenzie argues that they lack their own topic situation pronouns. In these cases, the intensional operator composes with the entire subordinate TP. Such clauses are *intensional* because a higher operator shifts their truth-conditional evaluation from the actual discourse situation to possible alternatives; it abstracts over the clauses’ situation arguments, binding them and turning the clauses into properties of situations. In these contexts, the subordinate clauses’ situation variables are left unsaturated (bound by the intensional operator), so SR^0 does not compose with a topic situation pronoun in the pivot clause and, instead, composes with the subordinate subject through function application. As seen in example (18) below, only canonical same-subject switch reference is felicitous in temporal subordinate clauses, even when the topic situations are distinct:

Kiowa

- (18) a. *Travis Carnegiecà tháuchě, hàun Énédăukòcà dăumâu.*
Travis Carnegie-kjæ Ø-tʰɔ:=tsě: hɔ̃n énédɔ:ko-kjæ Ø-dɔ:mɔ̃:
 T. C.-in 3s-stay=**when:SS** not Anadarko-in 3s-be-NEG
 ‘When Travis stays in Carnegie (SS), he isn’t in Anadarko.’
 b. # *Travis Carnegiecà tháuchě, hàun Énédăukòcà dăumâu.*
 ‘When Travis stays in Carnegie (DS), he isn’t in Anadarko.’ (McKenzie 2012:172)

As a result, under this analysis, languages are restricted to non-canonical switch-reference in non-intensional contexts and exclusively display canonical (i.e., subject-determined) switch-reference under intensional subordinators.

I follow McKenzie’s insight that every non-intensional clause contains a null topic situation pronoun, but depart from his implementation in two respects. First, rather than giving the switch-reference head inherent [SS] or [DS] features, I treat the SR^0 as bearing a Multiple Agree probe that copies index values of either subjects or topic situation pronouns (adapting the mechanism presented in Arregi & Hanink (2022)). Second, whereas McKenzie ties non-canonical switch-reference strictly to the absence of an intensional operator, I allow cross-linguistic variation—it depends on a parameter determining whether the probe in SR^0 targets topic situation pronouns or subjects in non-intensional contexts. This revised architecture preserves McKenzie’s key connection between switch reference and situation semantics to account for non-canonical switch reference, and accommodates cases, such as Hopi, where switch reference is always canonical, even in non-intensional contexts.

5.2 Switch reference as Multiple Agree

Whereas McKenzie (2012) explains non-canonical switch reference via topic situation pronouns, Arregi & Hanink (2022) focus on a canonical switch reference language and develop a Multiple Agree analysis to derive switch reference purely through a syntactic mechanism. The authors draw their data from Washo (Hokan/isolate), a language spoken along the California-Nevada border. Unlike Kiowa, the language examined by McKenzie (2012), there are no other complementizers in Washo apart from switch-reference morphology. Thus, the authors do not need to assume switch-reference morphology heads its own projection. Furthermore, there is only canonical switch reference in Washo. All switch-reference morphology tracks is whether subjects are referentially distinct or not. The authors propose that

switch reference results from a Multiple Agree operation between C^0 and the referential index value of the nominative-marked subjects in the pivot and anti-pivot clauses. Thus, embedded C^0 bears a probe for referential index features (labeled [ID]) that probes downward for the embedded subject's index features and probes upwards for the matrix subject index. At the end of the derivation, the result is that the probe in C^0 bears two copies of [ID], one for each subject. When [ID] in C^0 bears two distinct values (i.e., [_C ID:*i*, ID:*j*]), Vocabulary Insertion realizes the different-subject marker *š*, as seen in the following vocabulary entry:

$$(19)[_{C \text{ ID:}i, \text{ ID:}j}] \rightarrow \text{š} \quad (\text{where } i \neq j) \text{ DS (Arregi \& Hanink 2022:667)}$$

If the copied indices are identical, [ID] bears only one value (i.e., [_C ID: *i*, ID:*i*] = [_C ID: *i*]). Because the conditions for (19) are not met, the elsewhere rule in (20) applies, and C^0 is realized as the null suffix $\emptyset$:

$$(20)[_{C \text{ }}] \rightarrow \emptyset \quad \text{SS (Arregi \& Hanink 2022:667)}$$

The distributional properties of switch reference in Washo follow from a Multiple Agree analysis in which SR^0 carries an insatiable [ID] probe: it continues probing within its syntactic domain until it is exhausted. In practice, phase-boundedness limits that search. Because SR^0 is merged at the C-edge of the embedded CP, it can probe downward inside that CP to copy the embedded subject's index and probe upward one phase edge to copy the matrix subject's index, but it cannot cross an additional CP boundary. Therefore, at most two subjects mutually participate in the Agree relation, and a single switch-reference exponent is realized between two successive subjects. In the example below, there is no switch-reference marking for the simple reason that they are two distinct sentences not dominated by the same CP:

Washo

$$(21)[\text{pro}_i \text{ } \text{ʔlɔ:t} \quad \text{stɔr-a} \quad \text{l-éy-eweʔ-áyʔig-i}]$$

[*pro*_i yesterday store-OBL 1-go-hence-PST-IND]

[*gi*:-saʔ wáding^weʔé:beʔ **Ryan**_j ʔ-íy-eweʔ-ášaʔ-i]

[3:PRO-also today **Ryan**_j 3-go-hence-PROSP-IND]

'I went to the store yesterday. Ryan will go today, too.' (Arregi & Hanink 2022:659)

Unlike McKenzie (2012), however, there are no features [DS] and [SS] intrinsic to the head realizing switch reference and morphological insertion does not require an explicit semantic analysis. The eventual morphological insertion of switch reference is uniquely dependent upon the results of the index values derived through Multiple Agree with the relevant goals.

6. Switch reference in two Uto-Aztecan languages

In this section, I provide a comparative analysis of the switch-reference patterns of two Uto-Aztecan languages, Hopi and Serrano. I first examine Hopi's exclusively canonical switch-reference system, highlighting how switch-reference marking consistently tracks subject reference even in non-intensional contexts. Then, I discuss Serrano's non-canonical switch-reference system, illustrating how its morphology encodes situational continuity rather than subject identity. This juxtaposition lays the groundwork for my core proposal that both systems can be captured through parametric variation of the probe in SR^0 within a unified syntactic framework.

6.1 Only canonical switch reference: The case of Hopi

Hopi is a Northern Uto-Aztecan language spoken by the Hopi people in Northeastern Arizona. It constitutes its own subgroup within the Northern branch of Uto-Aztecan languages. It is characterized by unmarked SOV word order and a nominative-accusative case system (Jeanne 1978). In Hopi, switch-reference morphemes are clearly attached to the verb in the pivot clause as complementizers, and switch reference is always canonical (Hale 1992; Jeanne 1978). In (22), switch-reference marking, in addition to its primary function signaling non-coreferential subjects, carries particular temporal semantics as indicating a non-simultaneous temporal relation between the pivot and anti-pivot clauses. (23) shares the same temporal semantics, but its insertion is determined by the fact that the subjects of the pivot and anti-pivot clauses are coreferential:

Hopi

(22) *ʔi-pava_j paki-q (pii?) pam_k qatipti.*³

ʔi-pava_j paki-q (pii?) pam_k qatipti.

My-brother_j enter-DS (then) he_{k/*j} sat:down.

‘When my brother_j came in, he_{k/*j} sat down.’ (Jeanne 1978:183)

(23) *ʔi-pava_j paki-t (pii?) pam qatipti.*

ʔi-pava_j paki-t (pii?) pam_j qatipti

my-brother_j enter-SS (then) he_{j/*k} sat:down

‘When my brother_i came in, he_j sat down’ (Jeanne 1978:183)

In contrast to Serrano, however, Hopi does not limit switch-reference marking to temporal adverbial clauses (Hill 2016). In (24), it introduces a purpose relation between the pivot and anti-pivot clauses. In this example, it is also possible to see that switch-reference marking in Hopi can occur in complement clauses. In those cases, *-t* is the different-subject suffix, unlike in adverbial clauses, such as (23), where it is the same-subject suffix:

Hopi

(24) *Taaqa tiyo-t [pro kaway-mu-y ‘oyato-ni-qa-t] ‘ayata.*

Taaqa tiyo-t [pro kaway-mu-y ‘oyato-ni-qa-t] ‘ayata.

Man boy-ACC [(he) horse-PL-ACC put-FUT-C-ACC:DS] send

‘The man sent the boy to put the horses back.’ (Hale 1992:67)

The fundamental observation, however, about Hopi’s switch-reference morphology is that it consistently encodes whether the subjects of the pivot and anti-pivot clauses are coreferential or not—doing so without exception (Hale 1992; Jeanne 1978)—in contrast to non-canonical systems that mark whether the two clauses share the same topic situation or not. This conclusion is corroborated by the available Hopi documentation. Specifically, *Hopi Time*, Ekkehart Malotki’s 700-page seminal work published in 1983, refuting Whorf’s claim that Hopi lacks any notion of time, contains 552 instances of different-subject morphology and 340 instances of same-subject morphology. All of these instances of switch reference are instances of canonical switch reference. Hopi’s paradigmatic contrast between same-subject and

³ In this work, a single star * marks unacceptability and two stars ** marks a reconstructed form.

different-subject marking appears to be strictly defined in terms of subject identity. Table 1 presents the switch reference exponents in Hopi adverbial clauses; the set of clause-level relations is exhaustive for adjunct (adverbial) contexts. The temporal columns are particularly relevant here, because Serrano restricts switch reference to temporal contexts; the comparison and analysis that follows pivots on that shared semantic domain⁴:

Table 1: Hopi SR morphology (adapted from Hill 2016)

	Temporal Adverbial		Conditional 1	Conditional 2	Purpose	General
	Simultaneous	Non-Simultaneous				
SS	<i>-kyangw~- kyango</i>	<i>-t < *-tin</i>	<i>-nik(t)</i>	<i>-e'</i>	<i>-nen</i>	<i>-qe < *-ka</i>
DS	<i>-qw~-qö < *-ku</i>	<i>-qw~-qö < *-ku</i>	<i>-qw~-qö < *-ku</i>	<i>-qw~-qö < *-ku</i>	<i>-qw~-qö < *-ku</i>	<i>-qw~-qö < *-ku</i>

At the same time, the absence of non-canonical switch reference indicates that the relation between distinct situations is not the primary source of referential tracking for morphological exponence in Hopi's switch-reference system. Under the present proposal, this consistent correlation between subject identity and switch-reference marking follows from parametric variation in how SR^0 is configured to probe for referential index values ([ID]). Specifically, in Hopi, the probe in SR^0 exclusively targets nominative subjects, probing downward within the pivot clause and upward into the anti-pivot clause to copy each subject's referential index value. This parameterized behavior ensures that the morphology realized at SR^0 uniquely

⁴ Hopi also licenses switch-reference marking in relative clauses with the clause-level semantic interpretations associated with Table 1. In these cases, different-subject switch reference surfaces as *-t* and same subject switch reference surfaces as *-y*.

encodes whether the embedded subject's [ID] matches that of the matrix subject. Therefore, Hopi's switch-reference marking is invariably canonical. In other words, the syntactic parameter is set such that SR^0 ignores by Minimal Search non-subject elements, including null topic situation pronouns in Spec, TP, rendering morphological exponence independent of the identity of the situations of the two clauses.

This parametric option contrasts with that taken by languages like Serrano, which alter the parametric setting of SR^0 to probe for topic situation pronouns in non-intensional contexts. As I will show below, this change causes switch-reference morphology to correlate with the perceived unity or disunity of the two clauses' situation under discussion rather than with the identity of the subjects in the two clauses. Thus, Hopi and Serrano represent two sides of the same underlying switch-reference mechanism, distinguished by whether SR^0 is parameterized to target nominal subjects or topic situation pronouns in non-intensional contexts.

6.2 Only non-canonical switch reference: The case of Serrano

Unlike Hopi, Serrano licenses non-canonical switch reference. I propose that Austinian situation pronouns, rather than subjects, determine switch-reference marking in Serrano. Consequently, switch reference in Serrano encodes whether two or more clauses share the same topic situation rather than tracking subject co-reference between clauses. This analysis differs systematically from Hopi, in which switch reference uniquely marks subject identity.

Before exploring the particulars of Serrano switch-reference morphology, it is important to note that both Serrano and Hopi license clause chaining (Hill & Hill 2019; Toosarvandani 2024). Clause chaining refers to a syntactic configuration characterized by a series of verbs lacking the full range of inflectional marking. This sequence is then followed by a verb bearing said inflectional marking and typically specifying the temporal relation between the chained elements (Weisser 2013), which has been analyzed as an instance of asymmetric coordination

(Nonato 2014). In the case of Hopi, the final verb surfaces in the bare form, as Hopi lacks morphological tense (Toosarvandani 2024; Hale 1992). Clause chaining occurs in a non-intensional context that frequently bears switch-reference morphology and Merges distinct topic situation pronouns. On this basis, McKenzie (2012) identifies it as one of the only two configurations that license non-canonical switch reference. In this sense, it parallels coordinate structures introduced by an overt conjunction. Thus, one might hypothesize, following McKenzie (2012), that the presence of clause chaining exclusively in Serrano could explain the emergence of non-canonical switch reference in the language. However, Hopi also exhibits clause chaining, but remains a canonical switch-reference language, as I later show. Therefore, the difference between these two Uto-Aztec languages cannot be reduced to a simple distinction in the availability of clause chaining. Instead, we must look deeper into how the goal of the [ID] probe in SR⁰ is parameterized in each language.

In Serrano, switch-reference marking is exclusively associated with temporal semantics, expressing simultaneity or sequentiality. Table 2, given below, shows the switch-reference exponents of Serrano. As expected, based on the assumption that it is Merged in a non-nominal projection in the left periphery of the clause, switch-reference marking does not exhibit any type of nominal inflection (Hill & Hill 2019:1000):

Table 2: Serrano SR morphology (Hill 2016; Hill & Hill 2019)

	Temporal Adverbial	
	Simultaneous	Sequential
SS	<i>-ju' < *-yu</i> <i>-ivaju'</i> <i>-iva</i>	<i>-n(u)k ~ -nkw ~ -wnk</i>
DS	<i>-w~-ou~o</i> <i>< *-ku</i>	<i>-w~-ou~o < *-ku</i>

The sentences in (25) exemplify the simultaneous same-situation morpheme *ivaju'*, and (26) exemplifies the sequential same-situation morpheme *nk(w)*:

Serrano

(25) a. *Apin* ‘*acar*’*kin nepuhcay*’*veyu*’ (R&E 611)

Aa-pia=n *a^Rt\$a^R7-k-in* [*ny* *puhche-ivaju7*].

DIST-LOC=1SG>3SG tire-K-CAUS [1SG>PL care.for-SS.SIMUL]

‘I’m tired of taking care of you pl.’

b. *Mitavu*’ ‘*amatunga*’ ‘*ani*’ ‘*ervayt meme*’ *heermc nyiivayu*’ (R&E 178)

Mi=ta=vy-7 *ama-tunga7 ani* *oo^Rvai7t* *mymy7* [*hoo^Rm-ch*

DUB=IRR=3SG-PST DIST-INS COMP right.away die:CMP [shaman-ABS

ñi-ivaju7].

be-SS:SIMUL]’

‘Maybe it was because of that (the shamanic power) that she had died so soon, being a shaman.’ (Shamans do not live long.)

(26) ‘*Ama*’ *kwenevu*’ *hevek hiwnk neert Gabrrilyaanu*’ *xhiit* (R&E 48)

Ama7=kwyny-vy *hyvy7-k* [*hiu-nk* *nyy^Rh-t* *gabriljaanu7*

DIST=QUOT=3SG>3PL greet-K [see-SS.SEQ woman-ABS Gabrielino

hii-t.]

INDF-ABS

‘The Gabrielino (Tongva) woman greeted them when she saw them.’

The different-situation switch-reference marker *-w* (*-u* after a consonant, also transcribed as *-ou* in Hill (2016)) does not mark any distinctions in simultaneity or sequentiality:

Serrano

(27) a. *Pavay’pa*’*m huu*’*ow hakup muurk*. (R&E 840)

[*Pyveipa7=m huu7o-w*] Ø *hakup muur-k*.

[if=3PL fart-DS] 3SG very stink-K

‘When they (stinkbugs) fart it really stinks.’

b. *Acham=ta=ch* [*Ø pahi7-ko-w*] *mi-iv*.

1PL.PRO=IRR=1PL [3SG dawn-K-DS] go-FUT

‘We’ll go in the morning (when it dawns).’ (Hill & Hill 2019:1004)

Under the hypothesis that SR⁰ probes for topic situation pronouns in Serrano, switch reference does not necessarily track subject co-reference. This hypothesis is confirmed by the available Serrano documentation. In (28), the subjects of the two clauses are identical (i.e., both first-singular); however, the speaker conveys that there is no explicit connection, causal or

otherwise, between the situation described in the first clause and that described in the second clause by employing different-situation marking. Thus, one derives the intended reading where the event of going to the hospital does not entail the event of hearing some particular information.

Serrano

(28) *‘Ap xhaip hospitalpnu’ qacew ‘apya’nu’ mamc.* (R&E 196)

[Aa-p haii-p hospital-p=ny-7 qat\$o-w] aa-pia7=ny-7

[DIST-LOC INDF-LOC hospital-LOC=1SG-PST be.there-DS] DIST-LOC=1

maamt\$.

SG>3SG-PST hear

‘While I was in the hospital somewhere I heard it there.’

Contrastively, (29) shows that if a discourse entails a unified sequence, same-situation marking is used. In this case, the pivot surfaces with same-situation morphology because dying follows naturally from drinking the poisonous liquid:

Serrano

(29) *Kwa’c memek pa’nkw, keym.* (R&E 847)

Kwy7=t\$ mymy7-k [paa7-nkw], ky-j=m.

POT=2SG die-K [drink-SS:SEQ] say-IND=3PL

‘You could die after drinking it, they say.’

It is also possible to see same-situation marking in different-subject contexts, as evident in (30). None of the nominal arguments are coreferential. However, the situation described by the speaker is part of the situation of the pivot clause (i.e., whenever it is a good night, it is invariably the case that the stars shine and are nicely visible):

Serrano

(30) *Pavay’pa’ ‘a’ayec ‘atuuk nyiivayu’ kwa’cpe’ hye’ huu’m kukpa’now, chi’akow ‘enaac.* (R&E 430)

[Pyveipa7 a7ayy-t\$ atuuk ñi-ivaju7] kwy7=t\$=py

[if;when good-ABS night be-SS:SIMUL] POT=2SG=2>3PL

hiy7see huu7-m [ku~kpaa-7no-w], [chi7aa7-ko-w ynaat\$].

see:IMP star-PL [DUR~shine-ST-DS] [be.visible-K-DS nicely]

‘When it is a good night, you can see the stars as they shine and are nicely visible.’

(30) is problematic under McKenzie’s (2012) analysis of Kiowa. His account predicts that non-canonical switch reference cannot appear under the scope of an intensional subordinator, such as *if* or *when*. An intensional operator embeds a function from situations to truth values, which prevents switch reference from binding the same situation variable. This is due to a type mismatch. SRP is of type t and cannot be the input to an intensional subordinator. An intensional subordinator expects a function from situations to truth values (i.e., $\langle s, t \rangle$) and returns a function from situations to truth values of type $\langle s, t \rangle$, shifting the clause to a different situation. For the variable of type s to embed, it effectively requires switch reference not to bind it, defeating the semantics of switch reference. If *pyveipa7* glossed as ‘if; when’ in (30) were an intensional subordinator in Serrano, then, according to McKenzie (2012), we would expect canonical switch reference to surface in such contexts. Instead, we see non-canonical switch reference. Thus, we must investigate whether *pyveipa7* can be analyzed as a conjunction or a non-intensional morpheme in a clause-chaining configuration to preserve the observation in McKenzie (2012).

It is quickly possible to discard conjunction as a possible categorial analysis of *pyveipa7* because (30) would constitute a typological *unicum* in attested coordination strategies. Indeed, coordination of the form *Conjunction-S₁ S₂* is not attested in any known language (Haspelmath 2004).

Given that both Serrano and Hopi are noted in the literature as licensing clause chaining (Toosarvandani 2024), we must now examine how both languages instantiate it in the available documentation. Determining whether both languages do so is crucial in the present analysis because, alongside coordination, it is one of the contexts where McKenzie (2012) predicts non-canonical switch reference to surface. This prediction derives from the shared non-intensional nature of clause chaining and coordination, where distinct topic situation pronouns are Merged. Thus, the availability of clause chaining in Serrano, but not in Hopi, could account for the

presence of non-canonical switch reference in the former but not in the latter. Clause chaining, as noted above, involves a series of verbs that lack the full range of inflectional marking, followed by a final verb that bears inflection applying to the entire clause (Weisser 2013). In the case of Hopi, which lacks morphological tense, the final verb surfaces in the “absolute” form (Raible 2001). This area of Hopi documentation suffers from a great variety of idiosyncratic descriptive terminology. By *absolute form*, Raible (2001) appears to indicate the *bare* form of Hopi verbs, which are necessarily unmarked for morphological tense. Nonetheless, the same tense interpretation, derived through strategies other than morphological affixation, applies uniformly to all chained elements in the construction. To illustrate a non-Uto-Aztecan example for the sake of cross-linguistic verification, (31) is drawn from Mian (Papuan, Papua New Guinea) where the future tense and perfective aspect are expressed only on the final verb *ken* ‘do,’ but also apply to the medial verb *tl-* ‘come.’ Crucially, the medial verb is marked for switch reference, which is cross-linguistically common in clause chaining:

Mian

(31) *kóbó tl-ə-eb=ta okok ken-omab-bio=be*

you:SG come-DS:SEQ-2SG=MED work do:PERF-FUT-1PL=DECL

‘You’ll come and we’ll work’ (Weisser 2013:2)

Regarding the data available in Serrano, Hill & Hill (2019:991) note that the language has what they term “serial predicates.” I argue that this distinction is largely an issue of differing terminology and that Serrano has clause chaining. This claim leads to the prediction that there ought to be instances of a series of verbs unmarked for tense, aspect, modality, or evidentiality, followed by one final verb that exhibits some or all of these categories. This prediction is borne out in (32), where the two verbs *rapk* ‘become midday’ and *pynk* ‘pass’ are unmarked for any inflectional category and are both interpreted under the scope of the indicative-marked *ky-j* ‘say.’ Like (30), the example in (32) begins with a clause-initial temporal element, here *uvia* ‘already’ rather than *pyveipa7* ‘if; when’:

Serrano

(32) *Uvia hakup rap-k pyn-k, ky-j=n.*

Already very become.midday-*K* pass-*K* say-IND=1SG

‘It was already well past noon, I (mean to) say.’ (Hill & Hill 2019:991)

Because *pyveipa7* and *uvia* occupy the same left-edge slot, are followed by medial verbs that lack independent TAM marking, and display non-canonical switch reference, I suggest that the relevant segment in (30) can be considered as an instance of clause chaining. I treat (30) as a chaining construction in which *pyveipa7* contributes to the simultaneous interpretation of co-referential topic situations. However, it does not interfere with the Multiple Agree valuation that leads to the morphological exponence of same-situation switch reference.

In the case of Hopi, there are documented instances from Stahlschmidt (1983:521) (reported in Raible (2001)) of long sequences of chained medial verbs, each marked for switch reference, followed by a final verb in the absolute form:

Hopi

(33) *Noq yaw Tötölö-t nöm-'at nova-law-q*

and QUOT Grasshopper-SG:NOM wife-his-SG:NOM eat-DS

pam yaw nōma-y 'awinta-q yaw Isaw pasa-yamuy

he QUOT wife-his-SG.OBJ tell-DS QUOT Coyote field-their-PL.OBJ

aqle maknum-kyang put'aw pitu

near hunt-SS:SIMUL 3SG.ALLAT arrive-ABS.FORM

‘And while Grasshopper’s wife prepared the meal, he told her that, when Coyote was hunting on the fields close to them, he had approached them.’ (Stahlschmidt 1983:521)

One may legitimately wonder why the sequences in (33) warrant such a specific term as *clause chaining* when, at least superficially, they simply appear to contain instances of switch-reference marked coordinative morphology on verbs. However, it is precisely because of the interesting parallels between the description of clause chaining and the typology of coordination that analyses unifying the syntactic configurations of the two have been obtained (e.g., Weisser (2015); Nonato (2014)). These observations raise a broader question: if clause

chaining and asymmetric coordination share a common underlying structure, what does that imply for the cross-linguistic distribution and variation of non-canonical switch reference?

It is possible to see non-canonical switch reference in Serrano clause-chaining configurations. In (34) below, the medial verb *ñi* ‘do’ is marked with the same-situation suffix *ivaju*’ in a context where its subject and the subject of the final unmarked VP *vira*’n, *čaaçu*’ ‘speak (or) sing’ are disjoint in subject reference. Same-situation situation marking in (34) reflects the speaker’s “sense of an inevitable consequence between the clauses (as opposed to mere temporal consecutivity)” (Hill 2016:120):

Serrano

(34) *Taaqtam herheermim nyiivayu*’, ‘ama’ meempt kwenevu’ ‘apyu’ werra’n, *chaacu*’

(R&E 213)

[*Taaqt-am horhoorm-im ñi-ivaju*’], ama’ mĩĩmt kwini=vi’

[person-PL shamans-PL do-ss] that ocean QUOT=3SG>3SG

apiu’ *vira*’n, *čaaçu*’

from.there speak sing

‘If you are a shaman, the ocean would reportedly speak or sing (to you) from there; literally: ‘When people are shamans, that ocean can speak to him from there, (or) sing.’ (Hill 2016:120)

Since Serrano’s non-canonical switch reference also appears in clause-chaining configurations, it aligns well with McKenzie’s (2012) claim that clause chaining is one environment in which non-canonical switch reference is expected to occur.

However, an analysis that uniformly predicts non-canonical switch-reference marking in clause-chaining configurations crucially does not explain why Hopi, which also employs clause chaining, is a language with exclusively canonical switch reference. Thus, the ultimate distinction must lie in how the switch-reference probe is parameterized in non-intensional contexts in each language, rather than the sheer availability of clause chaining.

7. A diachronic reanalysis of switch reference: The case of Tohono O’odham

In the previous sections, I developed a parametric, Multiple-Agree analysis of switch-reference morphosyntax. In § 6.1, I argued that canonical switch reference in Hopi follows from a probe that targets nominative subjects. In § 6.2, I argued that parametrizing the same probe for topic situation pronouns derives the non-canonical pattern found in Serrano. These cross-linguistic contrasts suggest that the synchronic realization and parameter settings of switch-reference morphology are shaped by how child learners interpret the featural cues available in their input. A significant advantage offered by this parametric and acquisition-driven approach is that it makes clear predictions about synchronic variation and the types of diachronic paths switch-reference morphology can follow. Specifically, this approach leads to the prediction that when the primary linguistic input fails to provide clear cues to learners to hypothesize the referential-tracking feature of switch-reference morphology in non-intensional contexts, the switch-reference morphemes ought to be systematically reanalysed from one generation to the next. A promising setting to explore this diachronic prediction is Tohono O’odham (formerly known as Papago). Tohono O’odham is a Pimic Southern Uto-Aztecan language spoken in southern Arizona and northern Sonora, Mexico, that no longer has a productive switch-reference system. However, it retains unambiguous cognates of shared Uto-Aztecan switch-reference morphology (i.e., the different-subject suffix ***ku*, and the same-subject suffixes ***ka* and ***tin*). In contemporary O’odham, the descendant of Uto-Aztecan ***ku* surfaces as *ku-*, a morpheme coordinating two distinct clauses, obligatorily hosting a reduced version of the second clause’s auxiliary:

Tohono O’odham

(35) *’A’al ’o šoan kuc ’a:cim ñeñok.*

*’A’al ’o šoan **ku**=c ’a:cim ñeñok.*

Children AUX:IPFV:3PL cry:IPFV **KU**=AUX:IPFV:1PL 1PL speak:IPFV

‘The children are/were crying and we are/were speaking.’ (Zepeda 1983:27)

The cognates of Uto-Aztecan same-subject morphemes ***ka* and ***tin* respectively surface as the verbal suffixes *-k* and *-(k)c*. *-k* is used to conjoin verbs in the perfective form, marking

sequentiality between them (i.e., *and then*), and $-(k)c$ is used with verbs in the imperfective form, marking simultaneity, as noted in Mathiot (1974). The precise phonosyntactic characteristics of these morphemes are challenging to pin down from corpus sources because there are numerous instances where they surface following the direct object of a verb or where they are orthographically independent. Nonetheless, the latter occurs with perfective verbs and the former with imperfective verbs. Example (36) displays $-k$ as a verbal suffix between *m-ba'ăk* 'swallow you' and *m-wiñwua* 'regurgitate you.' Example (37) displays $(k)c$ as orthographically independent between two imperfective VPs:

Tohono O'odham

(36) *Şa ma:s mat ha'icu m-ba'ă-k m-wiñwua.*

Şa ma:s m=at ha'icu m-ba'ă-k

Rather. seem COMP=AUX:PFV:3SG something you-swallow:PFV-*K*

m-wiñwua.

you-regurgitate:PFV

"It rather looks like something swallowed you and spit you up (from your rumpled appearance)." (Hale 1983:304)

(37) *'A'al 'o cicwi kc hehem*

'A'al 'o cicwi kc hehem

children AUX:IPFV:3PL play:IPFV *KC* laugh:IPFV

'The children are/were playing and laughing.' (Zepeda 1983:24)

Thus, shared Uto-Aztecan switch-reference morphology persists in synchronic Tohono O'odham as coordinating morphology, varying in form depending on the syntactic category of the conjuncts. They no longer encode referential identity as a determining feature of their morphological exponence because they are no longer in complementary distribution; instead, they frequently occur in same- and different-subject contexts due to independent characteristics of the conjuncts they coordinate. In other words, in Tohono O'odham, the possibility of having either a single shared subject or two disjoint subjects is determined by the syntactic level at which coordination occurs. When coordination takes place inside the VP, Tohono O'odham

obligatorily interprets them with a single subject by virtue of the compositional properties of verb phrases in general; when two full clauses are coordinated, each clause bears its own independent subject. Thus, the distinct morphological forms presented in (35) – (37) are associated with joint or disjoint subjects by virtue of the structural position of their conjuncts. They do not presuppose conditions on subject identity for their morphological exponence that would qualify them as part of a productive switch-reference system. For example, the fact that *-(k)c* obtains a coreferential reading of the subjects of *cicwi* ‘play’ and *hehem* ‘laugh’ in (37) derives from the fact that coordination occurs at the VP level. All finite clauses in Tohono O’odham require an overt auxiliary, and there is no auxiliary-level coordination (Zepeda 1983; Johns p.c.). Moreover, in coordinated clauses, the subject of the second conjunct clause must be overt (Johns p.c.). Thus, the absence of a second auxiliary in (37) indicates that only one finite clause is projected, with a single external argument shared by both verbs. Therefore, coreference in (37) follows from VP-internal coordination, not switch-reference marking.

The central question is regarding which properties of Tohono O’odham morphosyntax can account for the loss of a productive switch-reference system and the reanalysis of Uto-Aztecan switch-reference morphology as coordinating particles between different syntactic categories. In this work, I argue that interaction between these morphosyntactic characteristics can be mediated by adopting Feature Economy as a driving force in diachronic reanalysis, as presented in Roberts & Roussou (2003) and Biberauer & Roberts (2017). This formal principle explicitly links the primary linguistic input to changes in the minimal featural representation that child learners must hypothesize to account for said input. In the next section, I introduce and explain Feature Economy and feature loss as mechanisms of diachronic change. I then turn to the relevant typological characteristics of Tohono O’odham that modified the input strings in such a way as to affect the minimal feature representation of Uto-Aztecan switch-reference morphology.

7.1 Theoretical background: Feature Economy and feature loss

The theoretical framework that I adopt to account for the reanalysis of Uto-Aztecan switch reference in Tohono O’odham is built upon the concepts of Feature Economy and feature loss, articulated by Roberts & Roussou (2003) and Biberauer & Roberts (2017). Under this framework, syntactic change is driven by differences in the feature representation of lexical items between the adult grammar and the one hypothesized by child learners. Having access only to the linear strings of adult output, learners acquiring a language posit only the minimal number of formal features in a hierarchical structure that are explicitly motivated by the surrounding linguistic input. In other words, a child learner encountering featurally ambiguous input hypothesizes fewer rather than more features, as long as this minimal set of features is sufficient to generate all observed strings in the environment.

This principle has significant repercussions for diachronic syntax. When the primary linguistic input no longer provides positive and explicit evidence for the presence of a particular formal feature, the child learner’s grammar differs from the adult’s because it does not hypothesize its existence, leading to feature loss.

7.2 The diachronic proposal

I propose that Tohono O’odham lost the switch-reference [ID] feature when three independent factors converged in the input available to a single generation of learners: (i) pervasive second-position auxiliaries agreeing with the subject’s ϕ -features, resulting in the acquisitional generalization of there being one auxiliary per well-formed clause; (ii) null-subject properties routinely removing the overt realization of coreferential anti-pivot subjects in same-subject marked clauses while maintaining the non-coreferential anti-pivot subject in different-subject marked clauses; and (iii) same-subject morphology attaching to pivot verbs in TAM-deficient environments. The resulting surface strings gave the child no opportunity to see the same-subject and different-subject markers in complementary distribution.

I propose three strings that capture, in turn, (i) the adult syntactic representation, (ii) an intermediate linear order before auxiliary cliticization, and (iii) the phonological output available to the child to account for the diachrony of the Uto-Aztecan same-subject markers. (38a) and (38b) represent the underlying syntactic organization of a same-subject marked clause in the adult grammar, where (38b) details an intermediate step prior to auxiliary cliticization for expository purposes. (38c) represents the phonological realization of the string in (38a) that serves as the primary linguistic input for the child learner:

(38) a. **[[SBJ_i ... V-κ/(κ)C] (SBJ_j) ... AUX V]

The inherited clause-chain layout: minimally inflected medial verb with a same-subject suffix, followed by a fully inflected clause.

b. **[[SBJ V-κ/(κ)C] AUX V]

The null-subject properties of O'odham delete the repeated subject of the second clause from the phonological realization

c. **[SBJ AUX V-κ/(κ)C V]

Auxiliary-second cliticization orders the auxiliary in second position. Thus, one auxiliary and normally one overt subject surface in the input string for the child learner. Thus, the subject of the subordinate/pivot clause is interpreted as the subject of the matrix/anti-pivot clause.

Across all three strings, the same-subject marker never appears beside a contrasting different-subject form. Thus, complementary distribution between these two forms does not materialize, and, by Feature Economy, the child learner hypothesizes the simplest representation consistent with the evidence and does not hypothesize an [ID] feature. I also propose that a parallel reconfiguration took place with different-subject *-ku, as presented below:

(39) a. **[[SBJ_i AUX₁ ... V-KU] SBJ_j ... AUX₂ V]

The initial different-subject layout, where both the pivot and the anti-pivot host an auxiliary and distinct subjects.

- b. $**[[\text{SBJ}_i \text{ AUX}_1 \dots \text{V-KU}] = \text{AUX}_2 \text{ SBJ}_j \dots \text{V}]$

Auxiliary-second order relocates AUX_2 leftward to cliticize onto the first constituent of the sequence. The input offers evidence of the $ku=\text{AUX}$ cluster at the juncture between the two clauses.

- c. $**[[\text{SBJ}_i \text{ AUX}_1 \dots \text{V}] [\text{KU}=[\text{AUX}_2 \text{ SBJ}_j \dots \text{V}]]]$

Learners reinterpret *ku-* as a clausal connector that obligatorily hosts AUX_2 .

Ku- links full clauses and obligatorily co-occurs with an overt second subject; as a result, it belongs to a syntactic environment distinct from the TAM-deficient, subject-optional domain where *-k/- (k)c* occur. The learner consequently never observes the two connectors in complementary distribution.

With no string that pairs two auxiliaries and two overt subjects around a potential switch-reference morpheme, the positive evidence for [ID] collapses. Feature Economy eliminates the feature, and Uto-Aztecan switch-reference morphology survives as coordinating particles. *Ku-* coordinates clauses; *-k* and *-(k)c* coordinate subclausal constituents with perfective-sequential and imperfective-simultaneous readings in the verbal domain, respectively.

7.3 Relevant properties of Tohono O’odham clauses

Three language-internal properties must be examined to explain why synchronic Tohono O’odham no longer provides learners with transparent evidence for an [ID] feature associated with switch reference. The first is obligatory auxiliary-second order in matrix and subordinate clauses. The second is null-subjects in matrix and subordinate clauses. The third, morphosyntactic deficiency in same-subject environments, emerges once we place the diachronic reconstruction in a broader typological context. The following subsections consider

each property in greater detail and show how they bleed the cues needed to hypothesize an [ID] feature, resulting in the loss of a productive switch-reference system.

7.3.1 Auxiliary-second order

With few exceptions, every clause, whether matrix or subordinate, in synchronic Tohono O'odham contains an auxiliary that cliticizes to the right edge of the first constituent. Every sentence can contain one and only one auxiliary (Zepeda 1983:19). Therefore, there is a general 1:1 correspondence between the number of auxiliaries and the number of clauses. The flexibility of Tohono O'odham word order results in a variety of different possible constituents surfacing in first position. For example, subject DPs (40a), object DPs (40b), verbs (40c), *wh*-phrases (40d), or complementizers (40e) can all be first-position constituents:

Tohono O'odham

(40) a. *Huan 'o wakon g ma:gina*

John AUX:IPFV:3SG wash:IPFV DET car'

b. *Ma:gina 'o wakon g Huan*

Car AUX:IPFV:3SG wash:IPFV DET John

c. *Wakon 'o g Huan g ma:gina*

Wash:IPFV AUX:IPFV:3SG DET John DET car

'John is/was washing the car.' (Zepeda 1983:130)

d. *Ša: 'o kaij g cehia?*

Ša: 'o kaij g cehia?

What AUX:IPFV:3SG say:IPFV DET girl

'What is the girl saying' (Zepeda 1983:53)

e. *'Añi 'añ cipkan wenog mo mumku g Huan.*

'Añi 'añ cipkan wenog m=o mumku

1SG AUX:IPFV:1SG work:IPFV while COMP=AUX:IPFV:3SG sick

g Huan.

DET John

'I am working while John is sick.' (Zepeda 1983:107)

This configuration has an important implication for the diachrony of switch-reference morphology because any historical switch-reference suffix that surfaced at the right edge of the

subordinate pivot clause became linearly adjacent to the auxiliary of the following clause once auxiliaries regularly cliticized to the first constituents' right edge. When this word-order fact is combined with a null-subject grammar (§ 7.3.2) and the inflectionally-deficient cross-linguistic environment of same-subject morphology (§ 7.3.3), positive cues for an [ID] feature disappear from the acquisitional input, supporting the Feature-Economy reanalysis presented in § 7.1.

7.3.2 Null subjects

Tohono O'odham freely omits pronominal subjects in both matrix and embedded clauses (Zepeda 1983:19), as seen in (41a) and (41b), respectively:

- Tohono O'odham
 (41) a. *Cicwi* 'o
 Play:IPFV AUX:IPFV:3SG
 'He/she is/was playing.' (Zepeda 1983:19)
- b. *B 'o kaij g Husi mañ g ma:gina eñga.*
 B 'o kaij g Husi m=añ
 PRT AUX:IPFV:3SG speak:IPFV DET Joe COMP=AUX:IPFV:1SG
 g ma:gina eñga.
 DET car own:IPFV
 'Joe said that I own a car.' (Zepeda 1983:108)

When coordinating clauses, however, the subject of the second coordinate clause must be overt, as we will see in detail when discussing the diachrony of different-subject morphology.

The null-subject profile of Tohono O'odham is relevant for the diachrony of switch reference because it systematically impoverishes the primary linguistic input of paired overt subjects. However, being a null-subject language is not in of itself enough to trigger the loss of a switch reference system. As presented in McKenzie (2012:176), Kiowa shows that a language can license null subjects and maintain a productive switch-reference system if same- and different-subject morphology continue to appear in complementary distribution. In the case of Tohono O'odham, this would entail same- and different-subject marking appearing alongside two overt

inflected auxiliaries. I argue that what was crucial in the development of same-subject morphology through the diachrony of Tohono O’odham is that null subjects converged with the general presence of clause chaining in Uto-Aztecan, in which medial verbs were deficient in TAM morphology and did not carry an auxiliary. In such strings, learners encountered a single auxiliary, at most one overt subject, and a connector in between the two verbs. Thus, when these factors are combined, the configuration that would have contained two fully inflected auxiliaries never surfaced with same-subject morphology; thus, the input offered no consistent cues to interpret the cognates of ***ka* and ***tin* as being in complementary distribution with the cognate of ***ku*. In the next subsection, I present the cross-linguistic evidence that same-subject morphology frequently occurs in morphosyntactically impoverished environments.

7.3.3 Morphological poverty in same-subject contexts

Unlike morphologically rich different-subject contexts (which I propose involved an auxiliary in each clause at an earlier diachronic stage of Tohono O’odham, clearly demarcating two distinct functional domains), same-subject marking typically exhibits a less morphosyntactically complex environment (Georgi 2012:4). Cross-linguistically, constituents marked with same-subject morphology often lack their own TAM morphology and are “dependent on the values of their reference clause that determines these values” (Georgi 2012:4). This typological generalization about morphosyntactic simplicity in same-subject contexts is particularly interesting when considered alongside observations about clause chaining. As seen above, clause chaining involves a series of minimally-inflected medial verbs, lacking TAM marking to varying degrees (Weisser 2013). For example, in Papuan languages exhibiting both clause chaining and switch reference, medial verbs surface with less inflectional marking than final verbs and cannot be used as independent units without the final TAM-marked verb (Georgi 2012:4). Similar morphosyntactic deficiency is also exemplified in

Inuktitut (Inuit–Yupik–Unangan, Canada), where same-subject-marked clauses “lack their own tense and are interpreted as simultaneous to the action of the reference clause” (Georgi 2012:5):

Inuktitut

(42) *Arna-p atisassat irrur-lu-git irinarsur-puq*

Woman-ERG clothes:ABS wash-SS-3PL sing-INTR:INDIC:3SG

‘While woman_i was washing the clothes, she_i sang.’ (Georgi 2012:6)

Thus, the typological picture offers a possible avenue to reconstruct the diachrony of Tohono O’odham, where the cognates of same-subject morphology became dissociated from their original subject-tracking function and were reanalyzed as conjunctions expressing temporal relations between verbs. Considered alongside the verb-second and null-subject properties outlined in §§ 7.3.1–7.3.2, these typological parallels remove the last overt cues that could sustain a contrastive and productive switch-reference system. The next section (§ 7.4) presents the diachronic account leading from canonical Uto-Aztecan same-subject switch-reference to the subclausal coordinators of modern Tohono O’odham.

7.4 The diachrony of same-subject morphology

Having presented the concomitant properties of Tohono O’odham that can account for the diachrony of switch-reference morphology, I now turn to their specific import in same-subject contexts. There are two relevant reconstructed Proto-Uto-Aztecan morphemes for the present discussion: ***-ka* and ***-tin*, whose descendants are present in a variety of Uto-Aztecan languages (e.g., Tümpisa Shoshone, Guarijio, Huichol) (Hill 2016:138). Tohono O’odham still preserves reflexes of these forms as the verbal suffixes *-(k)c* (cognate of ***-tin*) and *-k* (cognate of ***-ka*) and the generalized sub-clausal conjunction *(k)c* (also cognate of ***-tin*) (Mathiot 1974; Hale 1983). In this work, I focus on the development of these cognates in the verbal domain, exemplified by (43) and (44). In this context, they have been termed the *concurative* (i.e., *-k*) and the *correlative* (i.e., *-(k)c*) by Mathiot (1974). The examples in (43) are drawn

from Mathiot (1974), and the examples in (44) are drawn from Hale (1983). As noted in § 7.1, although the correlative *-(k)c* is clearly noted orthographically in the Mathiot dictionary as a verbal suffix, Hale (1983) provides attested instances of it being used to conjoin imperfective VPs following the first verb's object, as seen in (44b). The concursive *-k* is also attested conjoining perfective VPs following the first verb's object, as seen in (44a) :

Tohono O'odham

(43) a. *him-k*

go-*K*

'Having gone.' (Mathiot 1974:62)

b. *ali-kc*

child-*KC*

'Being a child.' (Mathiot 1974:62)

(44) a. *M 'ant o ha-ku'iwañ g şu:dagi k 'am o şul g mu:ñ.*

M 'ant o ha-ku'iwañ g şu:dagi k 'am

There AUX:PFV:1SG IRR OBJ-boil:CAUS:PFV DET water *K* there

o şul g mu:ñ.

IRR throw:PFV DET beans.

'I'll boil up some water and throw in some beans.' (Hale 1983:304)

b. *Ñ-'alidag 'o gegosid g gogs c ha-'i'icud g kakawyu.*

Ñ-'alidag 'o gegosid g gogs c ha-'i'icud g kakawyu.

My-child AUX:IPFV:3SG feed DET dog *C* them-water DET horses

'My kid feeds the dog and waters the horses.' (Hale 1983:305)

Because these morphemes no longer co-occur in complementary distribution with the synchronic descendants of different-subject morphology, Tohono O'odham cannot be said to have a productive switch-reference system. This outcome can be accounted for by the principle of Feature Economy, by which insufficient evidence in the input fails to trigger the hypothesis of an [ID] feature in the grammar of the acquiring child. Thus, same-subject marking is reanalyzed in the simplest way that can account for the primary linguistic input.

I argue that this reanalysis was triggered precisely by the loss of paradigmatic opposition between same-subject and different-subject morphology. Without a clearly contrasting

different-subject morpheme sharing the same distribution and providing overt evidence of differing options in subject tracking, the specialized [ID] feature associated with switch reference became opaque in the input. In the case of the synchronic cognates of same-subject marking, I propose that the fundamental ambiguity in the input arose from the presence of only one auxiliary in clause-chained configurations in a Proto-Pimic stage of the language, leading to two distinct morphosyntactic environments: one with only one overt auxiliary (i.e., same-subject contexts) and one with two overt auxiliaries (i.e., different-subject contexts). These considerations lead us to hypothesize a Proto-Pimic stage where the adult grammar retains a productive switch-reference system, but the underlying syntactic configuration of same-subject switch-reference clauses is influenced by TAM-deficient medial verbs and the language's licensing of null subjects. Example (45) below represents this underlying structure for the adult speaker prior to auxiliary-second cliticization:

(45) **[[(SBJ₁) V-*K*/(*K*)C] (SBJ₁) AUX V]

Because subject co-reference is necessary for the realization of same-subject morphology and the identity of the second occurrence of the subject is already given in the discourse, a language that licenses null subjects, like Tohono O'odham, frequently omits the overt realization of a repeated subject in a same-subject context. An intermediate string without an overt repetition of the subject is given below in (46):

(46) **[[(SBJ₁) V-*K*/(*K*)C] AUX V]

Null-subject configurations are not *a priori* incompatible with switch-reference morphology, as evidenced by Hopi and Serrano. In the case of Tohono O'odham, however, the defining factor was the joint effect of null-subjects and auxiliary-second cliticization inside same-subject configurations. In other words, if the child's input independently led to a robust and consistent generalization concerning second-position auxiliaries, then the same generalization could apply to the phonological realization of the string in (45). In the absence of a different-

subject morpheme in complementary distribution with same-subject marking in (46), no overt material triggers learning the [ID] feature associated with switch reference. Consequently, the suffix's original reference-tracking function remained opaque in the input, whereas its function of signaling the type of temporal relation between verbs is reinforced during acquisition. Under Feature Economy, the grammar that is retained is the one that treats the suffixes as subclausal conjunctions with particular temporal interpretations when associated with VPs. Thus, I propose that the application of auxiliary-second cliticization in same-subject marked clauses constituted the catalyst for variation between the grammars of different generations, leading us to the synchronic string seen in (47).:

(47) *[SBJ AUX V-*k*/(*k*)*c* V]

At the synchronic stage, the morphology no longer tracks subject co-reference as the determining factor in Vocabulary Insertion; instead, its distribution as a subclausal coordinator, limited to syntactic categories below the CP level, accounts for its surfacing in contexts where the understood subject of the marked and unmarked verbs is co-referential. The semantics of the two suffixes are associated with the temporal overlap between situations, or lack thereof, characterized by the perfective and imperfective aspects. Moreover, the correspondence between the number of auxiliaries and well-formed clauses highlighted in this section explains the unavailability of strings like (48): once a second auxiliary surfaces, O'dham shifts to the *ku*--based clausal-coordination strategy originating from different-subject switch reference. Sequences where two auxiliaries accompany *-k* or *-(k)c*, deriving coordination at the auxiliary level, are not part of the learners' input and remain unacceptable in contemporary Tohono O'dham (Johns p.c.):

(48) *[SUB AUX V CONJ AUX V]

The correspondence between well-formed clauses and auxiliaries sets up the diachronic trajectory of *ku*- explored in § 7.5.

7.5 The diachrony of different-subject morphology

Building on the account of the reanalysis of same-subject morphology presented in § 7.4, this section traces the diachronic path of the Uto-Aztecan different-subject suffix ***-ku* in Tohono O'odham. It shows how its interaction with the null subjects and auxiliary-second word order gave rise to the synchronic language's clausal-coordination strategy. I begin by illustrating the synchronic strategy for clause coordination, where we see the cognate surface as the optional prefix *ku-*. This prefix surfaces in between the two clauses hosting a phonologically reduced form of the second clause's auxiliary agreeing in person and number with the subject of the second clause (Zepeda 1983:26). As seen in the contrasting pair in (49), the deletion of the auxiliary's initial glottal stop and vowel constitutes the relevant phonological reduction. The first-person singular imperfective auxiliary *'aṇ* surfaces as *ṇ* when cliticized onto *ku-*. Finally, the subject of the second clause must be overt:

Tohono O'odham

(49) *'Uwĩ 'o cipkan kuṇ *('a:ṇi) 'o 'ohan.*

'Uwĩ 'o cipkan (ku)=ṇ

Woman AUX:IPFV:3SG work:IPFV (KU)=AUX:IPFV:1SG

**('a:ṇi) 'o 'ohan.*

1SG write:IPFV

'The woman is/was working, and I am/was writing.' (Zepeda 1983:27; Johns p.c.)

The same strategy holds in the perfective, where the third-person singular auxiliary *'at* surfaces as *t* when cliticized onto *ku-*:

(50) *Hegai 'uwĩ 'at 'am ṣonhi hegai ceoj kut 'am sosa.*

Hegai 'uwĩ 'at 'am ṣonhi hegai ceoj

That woman AUX:PFV:3SG LOC hit:PFV:SG that boy

ku=t 'am sosa.

KU=AUX:PFV:3SG LOC cry:PFV

'The woman hit the man and he cried (Scancarelli 1989:133)

Ku- may, however, be omitted. It is still the case that the auxiliary of the second clause surfaces in the same phonologically reduced form between the two clauses. In the example below, the second-person imperfective auxiliary *'ap* is reduced to *p*:

Tohono O'odham

(51) *Ceoj 'o ñeok p 'a:pi aš hehem.*

Ceoj 'o ñeok p 'a:pi aš hehem.

Boy AUX:IPFV:3SG speak:IPFV AUX:IPFV:2SG 2SG just laugh:IPFV

'The boy is/was speaking and you are/were just laughing.' (Zepeda 1983:26)

Interestingly, the third-person imperfective auxiliary in non-coordinated contexts is morphologically distinct from the other forms in the paradigm, surfacing as *'o*. However, it surfaces as *k* in coordination as a residue of *ku* after final-vowel deletion, as illustrated by (52):

Tohono O'odham

(52) *Kawyu 'o meḍ k hegai ceoj him*

Kawyu 'o meḍ k hegai ceoj him

Horse AUX:IPFV:3SG run:IPFV AUX:IPFV:3SG that boy walk:IPFV

'The horse is/was running and that boy is/was walking.' (Zepeda 1983:27)

In the following examples, I show how two separate clauses can be conjoined using the strategies outlined above. The (a) line in each example represents the sentences as they would appear independent of each other, and the (b) line represents their structure when coordinated:

Tohono O'odham

(53) a. *'Uwĩ 'o cipkan + A:ñi 'añ ko:š.*

'Uwĩ 'o cipkan + 'A:ñi 'añ ko:š.

Woman AUX:IPFV:3SG work:IPFV + 1SG AUX:IPFV:1SG sleep:IPFV

'The woman is/was working' + 'I am/was sleeping.'

b. *'Uwĩ 'o cipkan ñ 'a:ñi ko:š.*

'Uwĩ 'o cipkan ñ 'a:ñi ko:š

Woman AUX:IPFV:3SG work:IPFV AUX:IPFV:1SG 1SG sleep:IPFV

'The woman is/was working and I am/was sleeping.' (Zepeda 1983:25)

(54) a. *Ceoj 'o ñeok. + 'A:pi 'ap šoak*

Ceoj 'o ñeok + 'A:pi 'ap šoak

Boy AUX:IPFV:3SG speak:IPFV + 2SG AUX:IPFV:2SG cry:IPFV

‘The boy is/was speaking’ + ‘You are/were crying.’

b. *Ceoj ’o ñeok p ’a:pi şoak.*

Ceoj ’o ñeok p ’a:pi şoak

Boy AUX:IPFV:3SG speak:IPFV:SG AUX:IPFV:2SG 2SG cry:IPFV

‘The boy is/was speaking, and you are/were crying.’ (Zepeda 1983: 26)

Taken together, the coordination facts just reviewed—the one-auxiliary-per-clause pattern, the optional clause-initial *ku-* that hosts the reduced auxiliary of the second clause, and the morphologically distinct third-person coordinative auxiliary *k* in the imperfective—support a single diachronic scenario for different-subject marking. In the Proto-Pimic stage, the suffix ***ku* followed the pivot verb, explicitly signaling that the following clause would introduce a new subject, resulting in the configuration schematized below in (55):

(55) ***[[SBJ AUX₁ OBJ V-ku] [SBJ OBJ AUX₂ V]]*

With the emergence of obligatory auxiliary-second order, the auxiliary of the second clause (the anti-pivot, AUX₂) needed to appear in the second position within its own clause. The nearest available host was the first clause (with final *-ku*), leading AUX₂ to cliticize onto it, giving rise to the fused string schematized in (56):

(56) ***[[SBJ AUX₁ OBJ V-ku]=AUX₂ SBJ OBJ V]*

Once the *ku=AUX* cluster was linearly established, its original function of explicitly signaling subject discontinuity became obscured. Given the consistent SOV base order across Uto-Aztecan languages, the suffix *-ku* naturally stood at the right edge of the pivot clause, adjacent to the left periphery of the following clause. Because the auxiliary obligatorily raises to second position, the acquiring child encountered a single phonological cluster at this position that consistently correlated with a boundary between two clauses. Moreover, the loss of complementary distribution with same-subject morphology (§ 7.4), evident by the presence of two auxiliaries and an overt second subject, removed the evidence necessary to maintain a

productive switch-reference system. Under a Feature Economy-driven view of acquisition and diachronic change (Biberauer & Roberts 2017), the child had no incentive to posit an [ID] feature for a specialized switch-reference projection. Consequently, the learner retained only the semantic content recoverable from context, namely clausal coordination.

The synchronic distribution of *ku-* and other functional elements surfacing at clause junctures provides the evidence for the final step of the diachronic reanalysis. In contemporary Tohono O'odham, *ku-* consistently occupies the first constituent position, parallel to other clause-initial morphemes such as the complementizer *m-* 'that' and the polar-question particle *n-*, all of which trigger auxiliary-second order. Moreover, *ku-* may occur clause-initially before indefinite pronouns or adverbials in *wh-* questions, further aligning it with elements within the left periphery of the clause rather than with verbal morphology:

Tohono O'odham

(57) a. *Hegai 'uwĩ 'o s-ma:c map 'a:pi mumku.*

Hegai 'uwĩ 'o s-ma:c m=ap 'a:pi.

That woman AUX:IPFV:3SG knowing COMP=AUX:IPFV:2SG 2sg
mumku.

sick.

'That woman knows (is knowing) that you are sick.' (Zepeda 1983:108)

b. *No g mi:stol ko:š?*

N=o g mi:stol ko:š?

Q=AUX:IPFV:3SG DET cat sleep:IPFV:SG

'Is/was the cat sleeping?' (Zepeda 1983:14)

(58) *Kum hascu hihidoł 'a:pim?*

Ku=m hascu hihidoł 'a:pim?

KU=AUX:IPFV:2PL what cook:IPFV:PL 2PL

'What are/were you cooking?' (Zepeda 1983:55)

Given the presence in the input of various other functional elements at clause juncture, triggering auxiliary-second cliticization as the first constituent of their respective clauses, the

final step in the diachronic reanalysis resulted in the synchronic configuration where *ku-* was associated with the second clause in a similar fashion, as seen in (59):

(59) [[SBJ AUX₁ OBJ V] [KU [= AUX₂ SBJ OBJ V]]]

Crucially, contemporary learners do not encounter *ku-* in paradigmatic opposition with the diachronically same-subject markers *-(k)c* or *-k*, reinforcing the Feature Economy account whereby *ku-* is acquired solely as a coordinating morpheme hosting the auxiliary of the second clause. Thus, a suffix that originally tracked subject discontinuity evolved into a dedicated clausal coordinator marking clause boundaries rather than subject contrasts. This diachronic development is parallel to yet distinct from the reanalysis of same-subject morphology outlined in § 7.4.

8. Conclusion

In conclusion, I have argued that the typological diversity of switch reference can be accounted for by adopting a parametric analysis within the Multiple Agree framework originally presented by Arregi & Hanink (2022) to account for switch-reference in Washo. Under the present proposal, switch-reference morphemes are the realization of a specialized switch-reference projection bearing an [ID] feature whose goal is parametrized in non-intensional contexts: it can either probe and copy the referential index value of topic situation pronouns or that of subjects. Uniformly canonical switch-reference systems, such as the one in Hopi, derive from the latter parameter setting. Conversely, non-canonical switch reference systems, such as the one in Serrano, derive from the former.

This parametric split not only captures synchronic contrasts but also sheds light on the diachronic trajectories of switch-reference morphology. Under an acquisition-driven approach to diachronic change, Feature Economy can capture the loss of the [ID] feature associated with switch reference in languages such as Tohono O'odham, where the interaction of various syntactic innovations rendered the evidence of switch-reference markers in complementary

distribution opaque in the acquisitional input. Consequently, same-subject and different-subject morphemes were reanalysed as coordinating morphemes between specific syntactic categories.

Thus, this parametric Multiple Agree analysis has two clear advantages. First, it unifies canonical and non-canonical switch-reference patterns under a single syntactic mechanism, elucidating how limited parametric variation can systematically derive cross-linguistic differences. Second, by linking synchronic variation directly to acquisitional evidence, it provides a principled account of diachronic reanalysis and morphosyntactic change, as seen in Tohono O’odham. Overall, it simplifies the theoretical landscape and sets clear directions for further comparative research, language documentation, and diachronic investigation. Building on this foundation, future work could investigate whether the syntactic reanalysis outlined for Tohono O’odham can aid in clarifying the typological link between languages with clause-final inflected verbs and the presence of switch-reference morphology.

REFERENCES

- Ariel, Mira. 1990. *Accessing noun-phrase antecedents* (Routledge Library Editions: Linguistics), vol. 12. London & New York: Routledge.
- Arregi, Karlos & Hanink, Emily A. 2022. Switch reference as index agreement. *Natural Language & Linguistic Theory* 40(3). 651–702. (doi:[10.1007/s11049-021-09521-2](https://doi.org/10.1007/s11049-021-09521-2))
- Baker, Mark C. & Souza, Livia Camargo. 2020. In Barrie, M., Gillon, C., Haugen, J., Mathieu, E. & Siddiqi, D. (eds.) Switch-Reference in American Languages: A Synthetic Overview. *The Routledge Handbook of North American Languages*. London & New York: Routledge. (<https://doi.org/10.4324/9781315210636>)
- Biberauer, Theresa & Roberts, Ian. 2017. Parameter Setting. *The Cambridge Handbook of Historical Syntax*, 134–162. Cambridge: Cambridge University Press. (doi:[10.1017/9781107279070.008](https://doi.org/10.1017/9781107279070.008))
- Comrie, Bernard. 1983. Switch-reference in Huichol: A typological study. In Haiman, J. & Munro, P. (eds.), *Switch Reference and Universal Grammar: Proceedings of a symposium on switch reference and universal grammar, Winnipeg, May 1981* (Typological Studies in Language; vol. 2). Philadelphia: John Benjamins Publishing Company. (doi:[10.1075/tsl.2.04com](https://doi.org/10.1075/tsl.2.04com))
- de Sousa, Hilário. 2016. Some non-canonical switch reference systems and the fundamental functions of switch reference. In van Gijn, R. & Hammond, J. (eds.), *Switch Reference 2.0* (Typological Studies in Language; vol. 114), 55–92. Philadelphia: John Benjamins Publishing Company. (doi:[10.1075/tsl.114.02des](https://doi.org/10.1075/tsl.114.02des))
- Georgi, Doreen. 2012. Switch Reference by Movement. In Weisser, P. (ed.), *Perspectives on Switch-Reference: Local Modeling and Empirical Distribution*. (Linguistische Arbeits Berichte; vol. 89). 1–40. Leipzig.
- Haiman, John & Munro, Pamela. (eds.). 1983. *Switch Reference and Universal Grammar: Proceedings of a symposium on switch reference and universal grammar, Winnipeg, May 1981* (Typological Studies in Language; vol. 2). Philadelphia: John Benjamins Publishing Company.
- Hale, Ken. 1983. Papago (k)c. *International Journal of American Linguistics* (Papers Presented at a Symposium on Uto-Aztecan Historical Linguistics (Jul., 1983)). 49(3). 299–327.
- Hale, Ken. 1992. Subject Obviation, Switch Reference, and Control. In Larson, Richard K. & Iatridou, S. & Lahiri, U. & Higginbotham, J. (eds.), *Control and Grammar*, 51–77. Dordrecht: Springer Netherlands. (doi:[10.1007/978-94-015-7959-9_2](https://doi.org/10.1007/978-94-015-7959-9_2))

- Haspelmath, Martin. 2004. Coordinating constructions: An overview. In Haspelmath, M. (ed.), *Coordinating constructions*, 3-39. Amsterdam: John Benjamins Publishing Company (<https://doi.org/10.1075/tsl.58.03has>)
- Hill, Jane H. 2016. Takic switch reference in Uto-Aztecan perspective. In van Gijn, R. & Hammond, J. (eds.), *Switch Reference 2.0* (Typological Studies in Language; vol. 114), 115–152. John Benjamins Publishing Company. (doi:[10.1075/tsl.114.04hil](https://doi.org/10.1075/tsl.114.04hil))
- Hill, Jane H. & Hill, Kenneth C. 2019. *Comparative Takic grammar*. (<https://escholarship.org/uc/item/6tr732gg>)
- Jeanne, LaVerne Masayesva. 1978. *Aspects of Hopi Grammar*. Boston: Massachusetts Institute of Technology. (Doctoral Dissertation.) (<https://dspace.mit.edu/handle/1721.1/16325>)
- Malotki, Ekkehart. 1983. *Hopi time: A Linguistic Analysis of the temporal concepts in the Hopi language* (Trends in linguistics. Studies and monographs 20). Reprint 2011. Berlin & New York: Mouton Publishers.
- Mathiot, Madeline. 1974. *A Dictionary of Papago Usage*. Bloomington: Indiana University Press
- McKenzie, Andrew Robert. 2012. *The Role of Contextual Restriction in Reference-Tracking*. Amherst: University of Massachusetts. (Doctoral Dissertation) (<https://hdl.handle.net/20.500.14394/39025>)
- Nonato, Rafael. 2014. *Clause chaining, switch reference and coordination*. Boston: Massachusetts Institute of Technology. (Doctoral Dissertation) (<https://dspace.mit.edu/handle/1721.1/87499>)
- Ramón, Dorothy & Elliott, Eric. 2000. *Wayta' yawa' Always believe*. Banning: Malki-Ballena Press.
- Roberts, Ian & Roussou, Anna. 2003. *Syntactic Change: A Minimalist Approach to Grammaticalization* (Cambridge Studies in Linguistics). Cambridge: Cambridge University Press. (doi:[10.1017/CBO9780511486326](https://doi.org/10.1017/CBO9780511486326))
- Raible, Wolfgang. 2001. Linking clauses. In *Language Typology and Language Universals. An International Handbook*. Haspelmath, M., König, E., Oesterreicher, W., & Raible, W. (eds.), 590–617. Berlin & New York: Mouton de Gruyter
- Stahlschmidt, Andrea. 1983. Das Verbsystem des Hopi. Eine semantische Strukturanalyse der Hopi-Grammatik unter besonderer Berücksichtigung von B. L. Whorfs Thesen zur Zeitauffassung der Hopi-Indianer. *SAIS Arbeitsberichte aus dem Seminar für Allgemeine und Indogermanische Sprachwissenschaft*, 7. Kiel: Universität Kiel

- Stirling, Lesley. 2009. *Switch-reference and discourse representation* (Cambridge Studies in Linguistics; vol. 63). Cambridge: Cambridge University Press.
- Toosarvandani, Maziar. 2024. Clause chaining in Uto-Aztecan: A Northern Paiute perspective. In Sarvasy, H. S. & Aikhenvald, A. Y. (eds.), *Clause Chaining in the Languages of the World*. Oxford: Oxford University Press. (doi:[10.1093/oso/9780198870319.003.0012](https://doi.org/10.1093/oso/9780198870319.003.0012))
- Weisser, Philip. 2013. *Dissecting clause chaining constructions*. Presented at ConSOLE XXI. Potsdam
- Zepeda, Ofelia. 1983. *A Tohono O'odham grammar*. Tucson: University of Arizona Press.